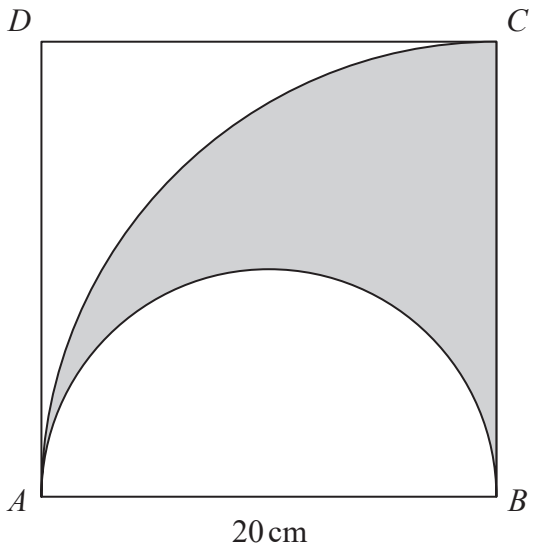
26 The diagram shows a square $ABCD$ with sides of length 20 cm.
It also shows a semicircle and an arc of a circle.



AB is the diameter of the semicircle.
AC is an arc of a circle with centre B.

Show that $\dfrac{\text{area of shaded region}}{\text{area of square}} = \dfrac{\pi}{8}$

Semicircle: $r = 10$ cm.

Whole area $= \pi \times 10^2 = 100\pi$.

	Angle	Area cm²	
÷2	360	100π	÷2
	180	50π	

1 mark
Find the semicircle area

Quarter-circle: $r = 20$ cm.

Whole area $= \pi \times 20^2 = 400\pi$.

	Angle	Area cm²	
÷4	360	400π	÷4
	90	100π	

1 mark
Find the quarter-circle area

Shaded area $= 100\pi - 50\pi = 50\pi$ cm².

Square area $= 20 \times 20 = 400$ cm².

1 mark
Subtract the semicircle and find the square area

Required fraction $= \dfrac{50\pi}{400} = \dfrac{\pi}{8}$

1 mark
Exact simplified result

(Total for Question 26 is 4 marks)